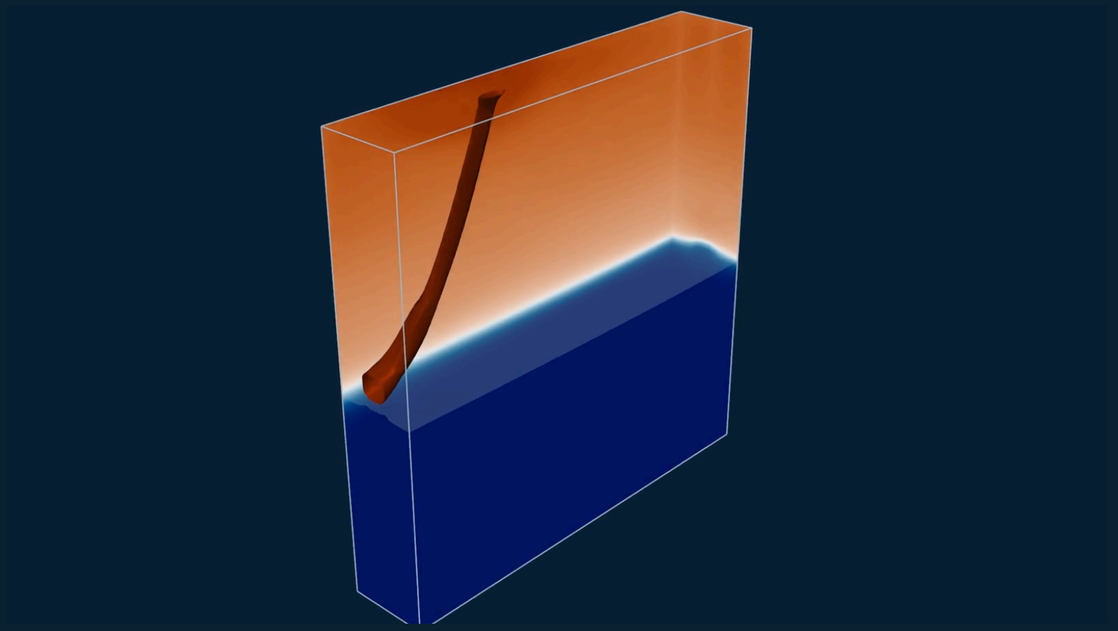
MASTER'S THESIS · AI FOR SIMULATION · JKU LINZ

# Can a recorded flow carry a temperature that *pushes back*?

Data-based Numerical Simulation of Buoyant Jet Flow



reference simulation, adiabatic tank, 3D render of stored solver states

OBSERVE

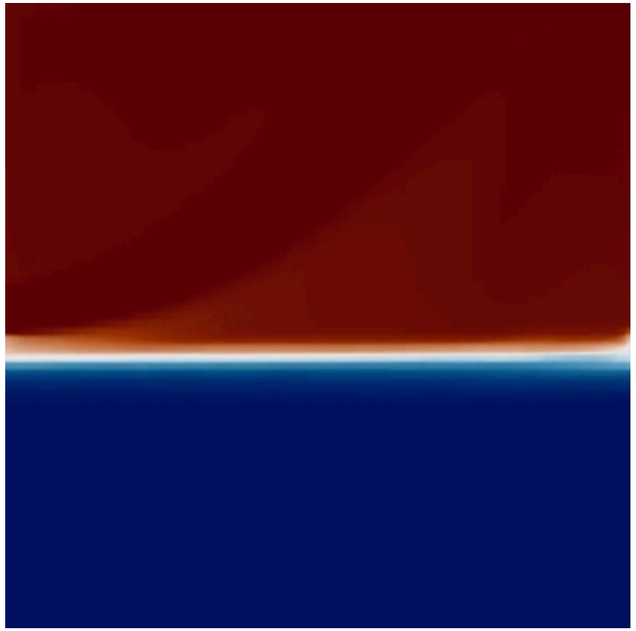
# A warm jet fills the top of the tank and builds a front only *five millimetres* thick.

**Warm inflow** at mid-height for 315 s, then cold inflow until 600 s

**Laminar**: Reynolds number about 100, no turbulence model

**Closed tank**, 75 mm: every joule in and out is accounted for

The **outlet** sits just below the front



reference CFD

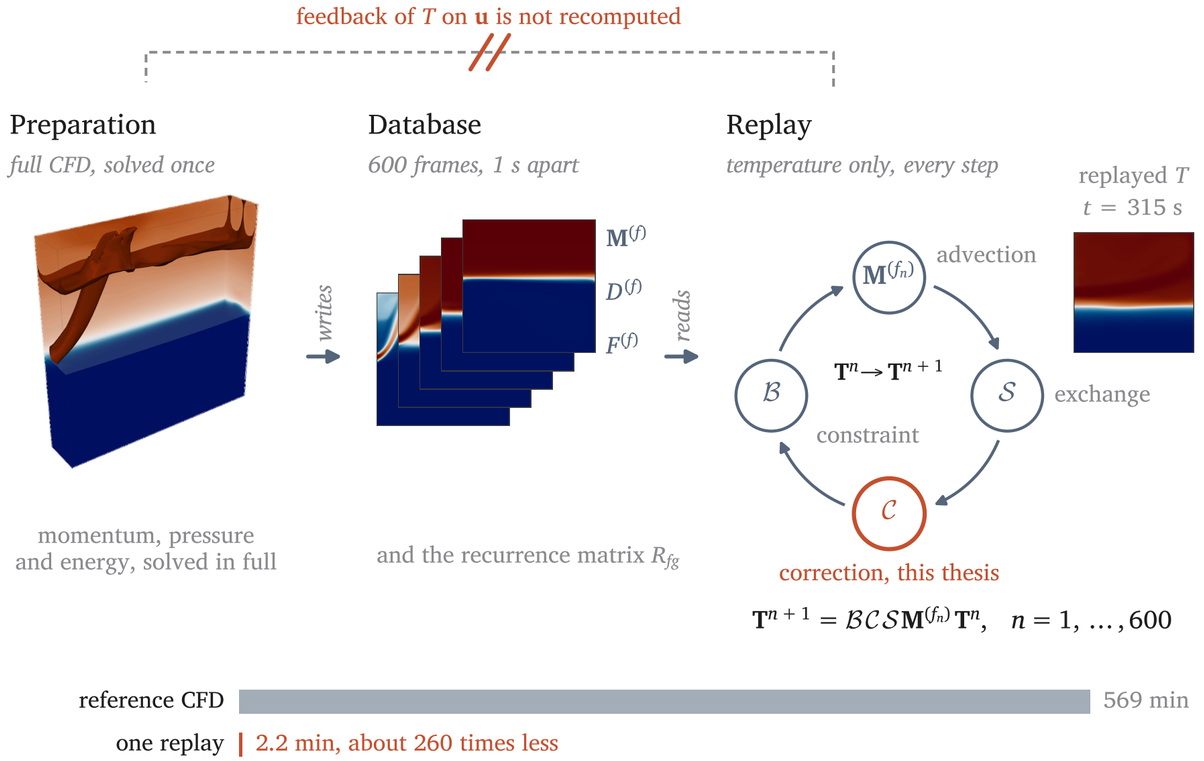
mid-plane temperature, 293-333 K, decoded from the rendered frames (mean error ≤ 0.5 K)

THESIS CH 5

293 K 313 K 333 K

METHOD

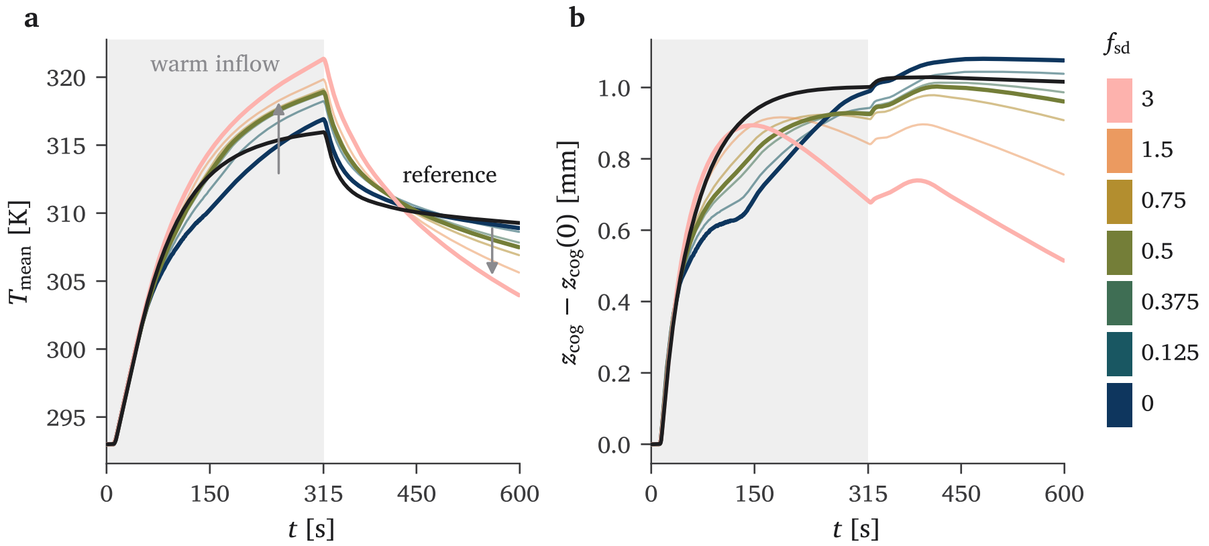
# Recurrence CFD records the motion once and replays it about *260 times* cheaper.




solver time without image output: reference 569 min, mean replay 2.16 min (range 120-700x); the recording itself is one full run

THESIS FIG 2.1 · SEC 5.1

# One mixing knob cannot get heat content and layer height right *at the same time*.




mean temperature and rise of the centre of gravity, seven strengths of the face exchange

THESIS FIG 5.1

**1.07 K**

best error of the mean temperature, with no added mixing ($f_{sd} = 0$)

**0.068 mm**

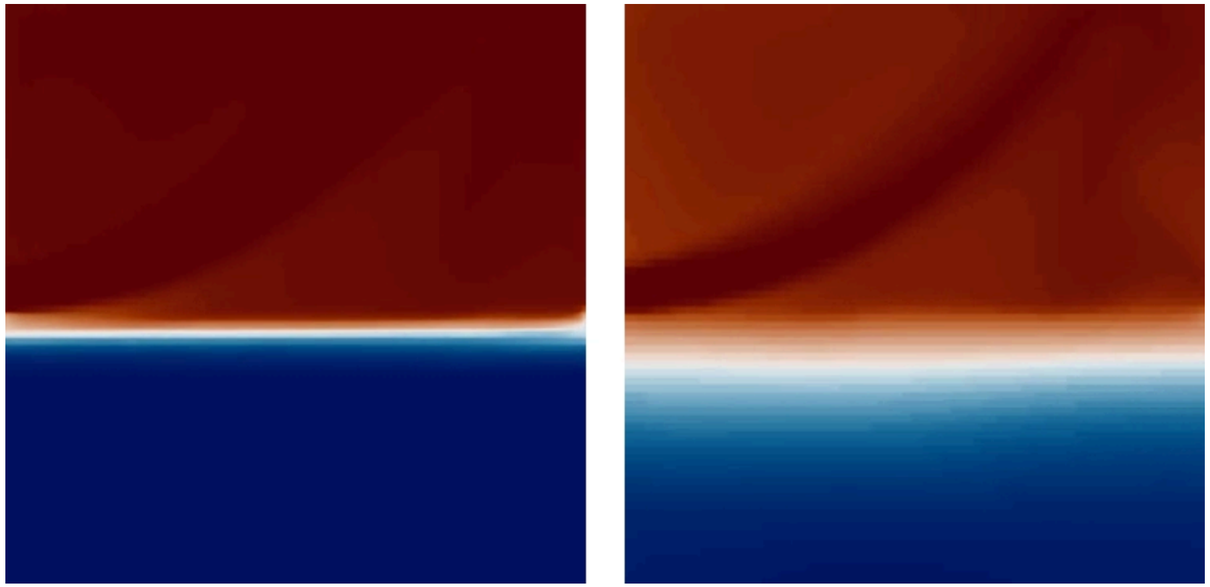
best error of the layer height, near $f_{sd} = 0.5$

**5.08 K**

too cold at the end with the strongest mixing, $f_{sd} = 3$

EXPLAIN · FROM THE TANK BUDGET DOWN TO ONE FACE

# The exchange smears the front, and the outlet below it *drains the heat*.



reference CFD

standard replay, $f_{sd} = 0.5$

mid-plane temperature, 293-333 K, decoded from the rendered frames (mean error ≤ 0.5 K)

THESIS CH 5

293 K 313 K 333 K

**4.5**
mm, reference front at 240 s

**19.3**
mm, standard replay

**2.5 mm**
with the recorded maps alone: the maps do not smear it

**4.6 K**
of mean temperature drained through the outlet after the inflow (bulk balance, Fig 5.10)

Q2 · THE CORRECTION OF THIS THESIS

# A *bounded* anti-diffusion rebuilds the front without creating new extremes.




model front: 40 K across 60 cells, ν = −0.1 against limited s = 0.25 (not replay data); shaded: outside the data range

THESIS FIG 4.1

Each face moves heat **up** the gradient, but only as far as both cells stay inside the range of their neighbours:

$$\Delta q_f = s \cdot min(Q^+{}_h, Q^-{}_c)$$

A local maximum cannot grow, a minimum cannot sink, and heat is conserved face by face.

Faces are weighted towards the normal of the front, here the vertical, so ripples along the front are not amplified.

Limiter in the spirit of flux-corrected transport (Boris & Book 1973; Zalesak 1979).

COMPARE · ADIABATIC TANK

# In the adiabatic tank the front returns to the *width of the reference*.

**15.7 mm**

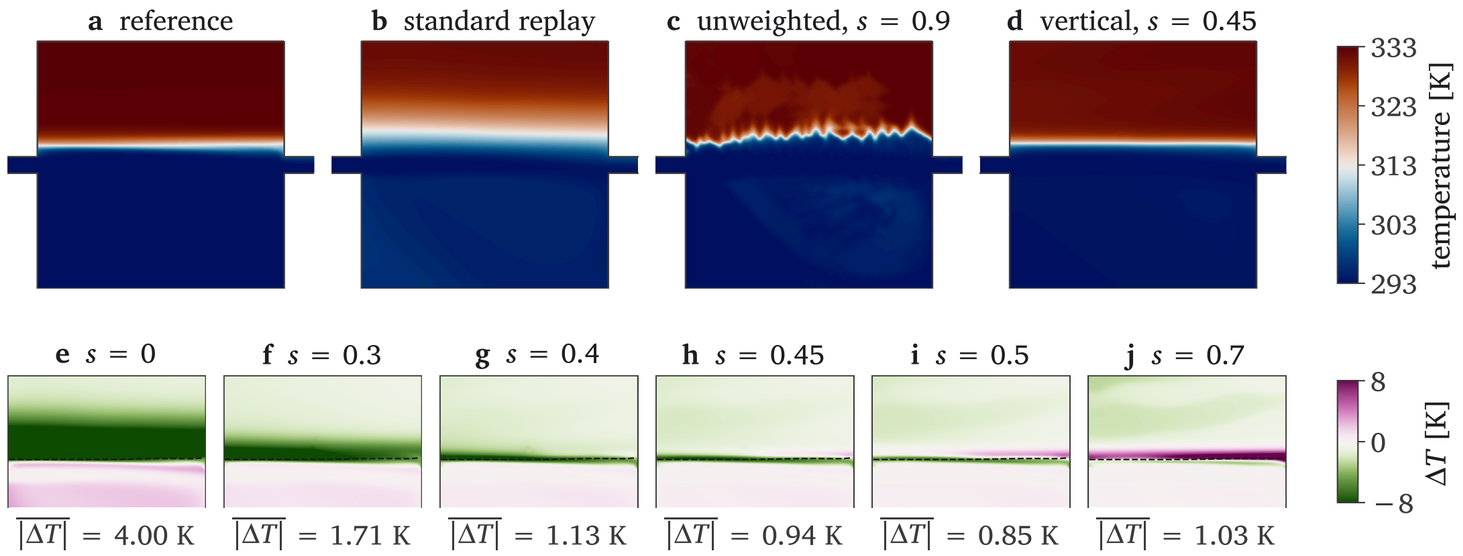
standard replay, end front

**5.3–4.5 mm**

corrected, working range s = 0.40–0.50; reference 5.0 mm

**1.46 → 0.70–0.64 K**

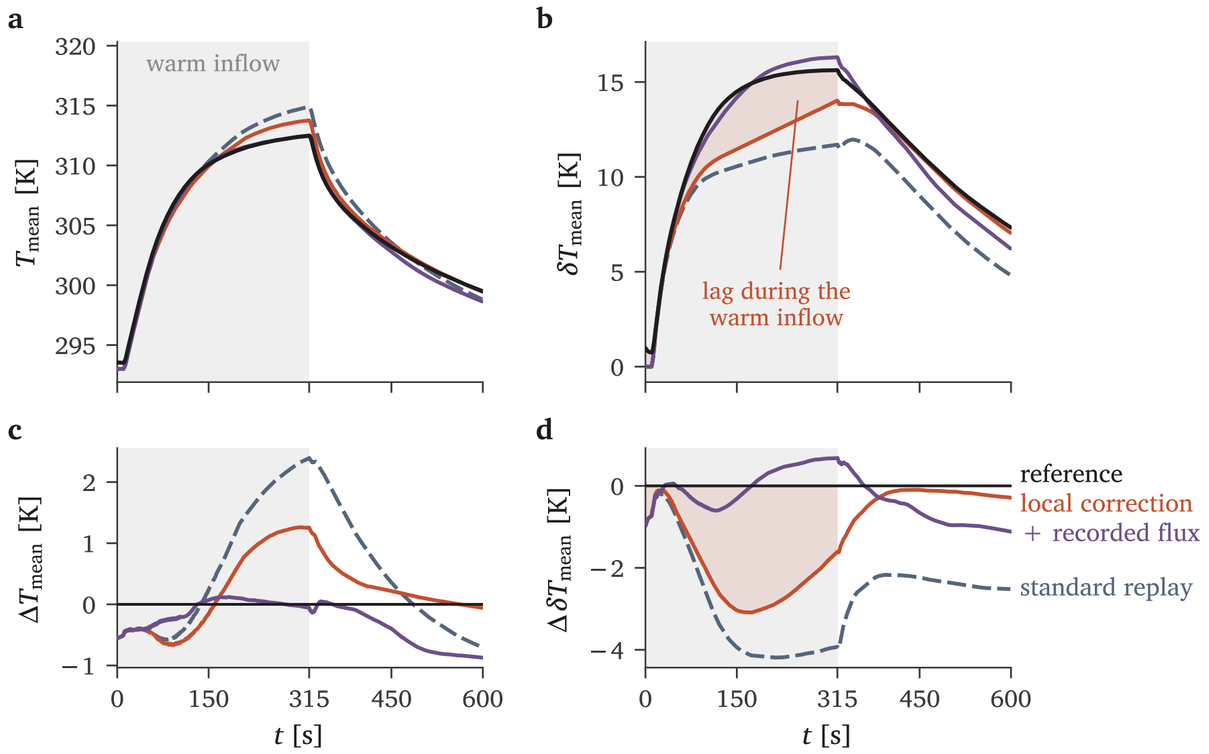
error of the mean temperature over the run




end state at 599 s and the strength ladder as departure from the reference; s was chosen by comparison with this reference

THESIS FIG 5.9

# Under a cooled lid the heat comes out right, but the layers *separate too slowly*.




mean temperature, heterogeneity and their departures; band: lag of the local correction while the tank fills

THESIS FIG 6.3

**−0.66 → +0.06 K**

end error on 9.62 K of cooling: standard replay, then the correction at $s \approx 0.55$

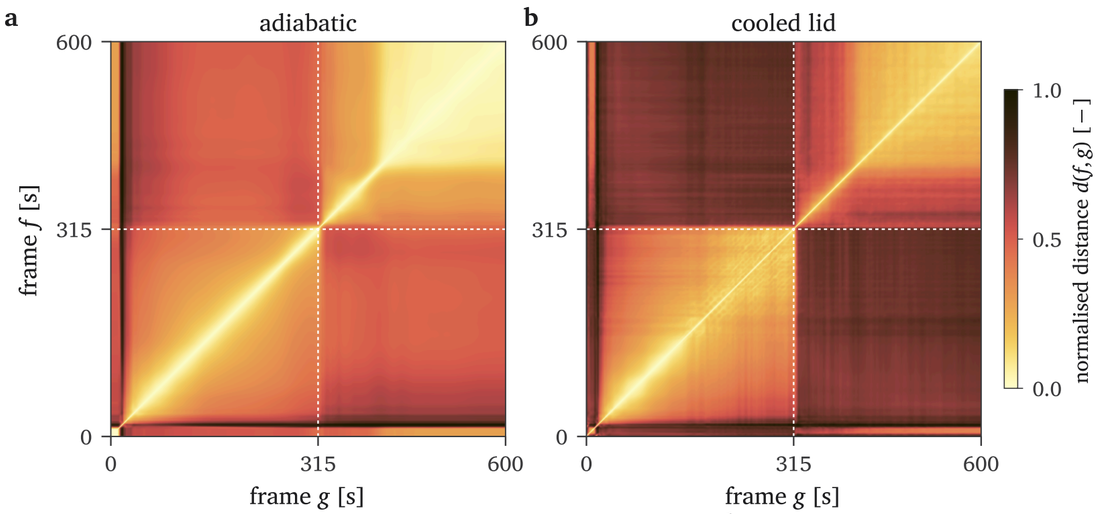
**2.25 → 0.45 K**

lag of the heterogeneity while filling, without and with a recorded vertical heat flux

**−0.86 K**

the price: the tank then ends too cold

# A recording belongs to the *flow regime* it was made in.




(c) after the inflow the cooled-lid flow renews about ten times faster at short lags; (d) neither recording recurs in the method's own terms

THESIS FIG 2.2

EVIDENCE AND LIMITS

# What the evidence supports, and what it *does not*.

SHOWN no single $f_{sd}$; the drain at the outlet is the cause

SHOWN a conservative, bounded correction rebuilds the front

SHOWN under cooling, the remaining error is missing transport

SHOWN a database belongs to its flow regime

OPEN a strength chosen without the reference

OPEN other meshes, turbulent or multiphase flows

OPEN the through-flow vessel (Q5): runs in progress

OPEN a recorded flux that transfers to other cases

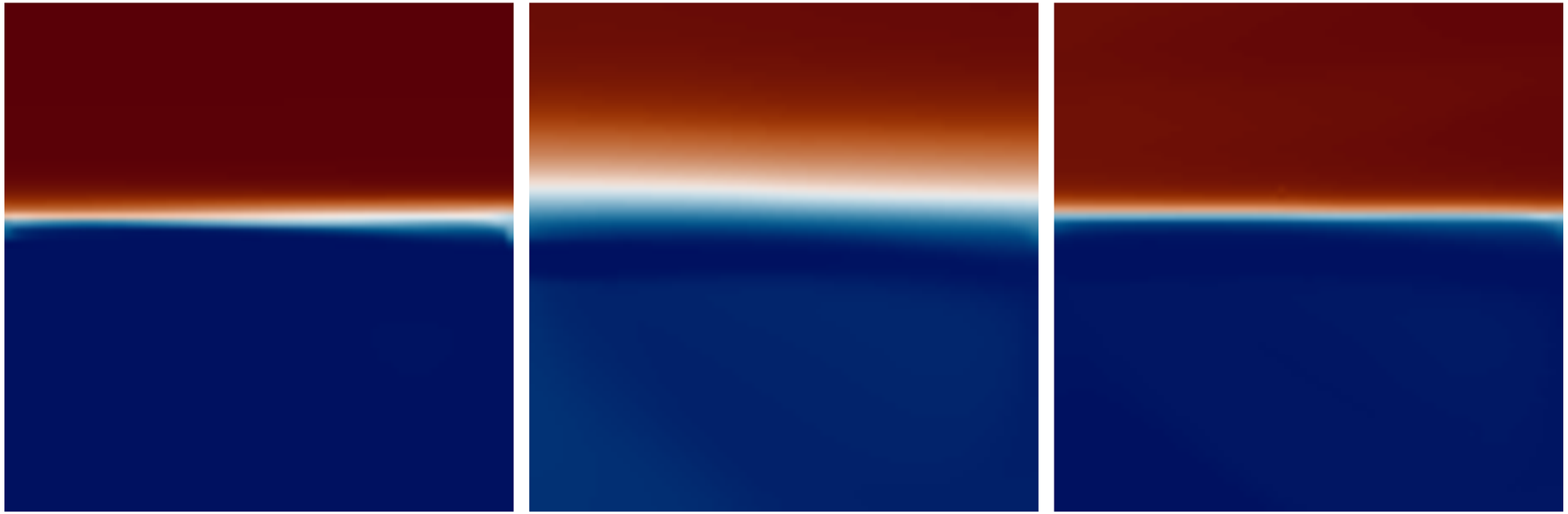
CONTRIBUTION

1. A measured trade-off in the standard replay, and its cause: a smeared front drains heat through the outlet.
2. A conservative, bounded anti-diffusive correction that rebuilds the front without new extremes.
3. The limit of local corrections: what remains is transport, and a recording belongs to its regime.

Thank you.

BACKUP · FRAME INSPECTION

# Stored snapshots: reference, standard and corrected replay.



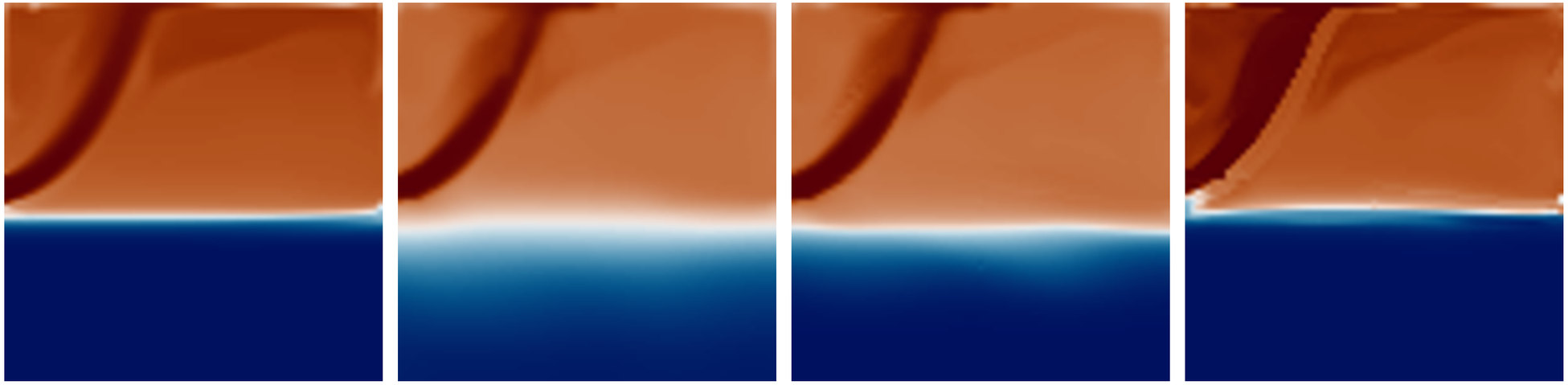
reference CFD | standard replay | corrected, $s = 0.45$

mid-plane temperature, 293-333 K; solver snapshots of the replays, reference states interpolated in time

STORED SOLVER SNAPSHOTS

BACKUP · FRAME INSPECTION

# Cooled lid: the recorded flux restores the layering while the tank fills.



reference CFD | standard replay | corrected, s = 0.475 | + recorded flux

cooled lid, h = 100; same scale 293-333 K

STORED SOLVER SNAPSHOTS

BACKUP · DERIVATION (THESIS APP. D)

# Below $s = 1/6$ no cell can leave its bounds, even before the clamp.

Each face gives at most $s \cdot Q^{+}_{h}$ to the warm cell and takes at most $s \cdot Q^{-}_{c}$ from the cold one.

A cell with $n_f$ faces receives at most $n_f s \cdot Q^{+}$ and gives at most $n_f s \cdot Q^{-}$.

So $-Q^{-} \le C \cdot \Delta T \le Q^{+}$ whenever $n_f s \le 1$, i.e. $\mathbf{s \le 1/6}$ on the hexahedral mesh.

Above 1/6 the clamp may act; the working range 0.40–0.50 relies on the clamp and the global conservation constraint.

BACKUP · COST

# 260 times is solver time against solver time, without the recording.

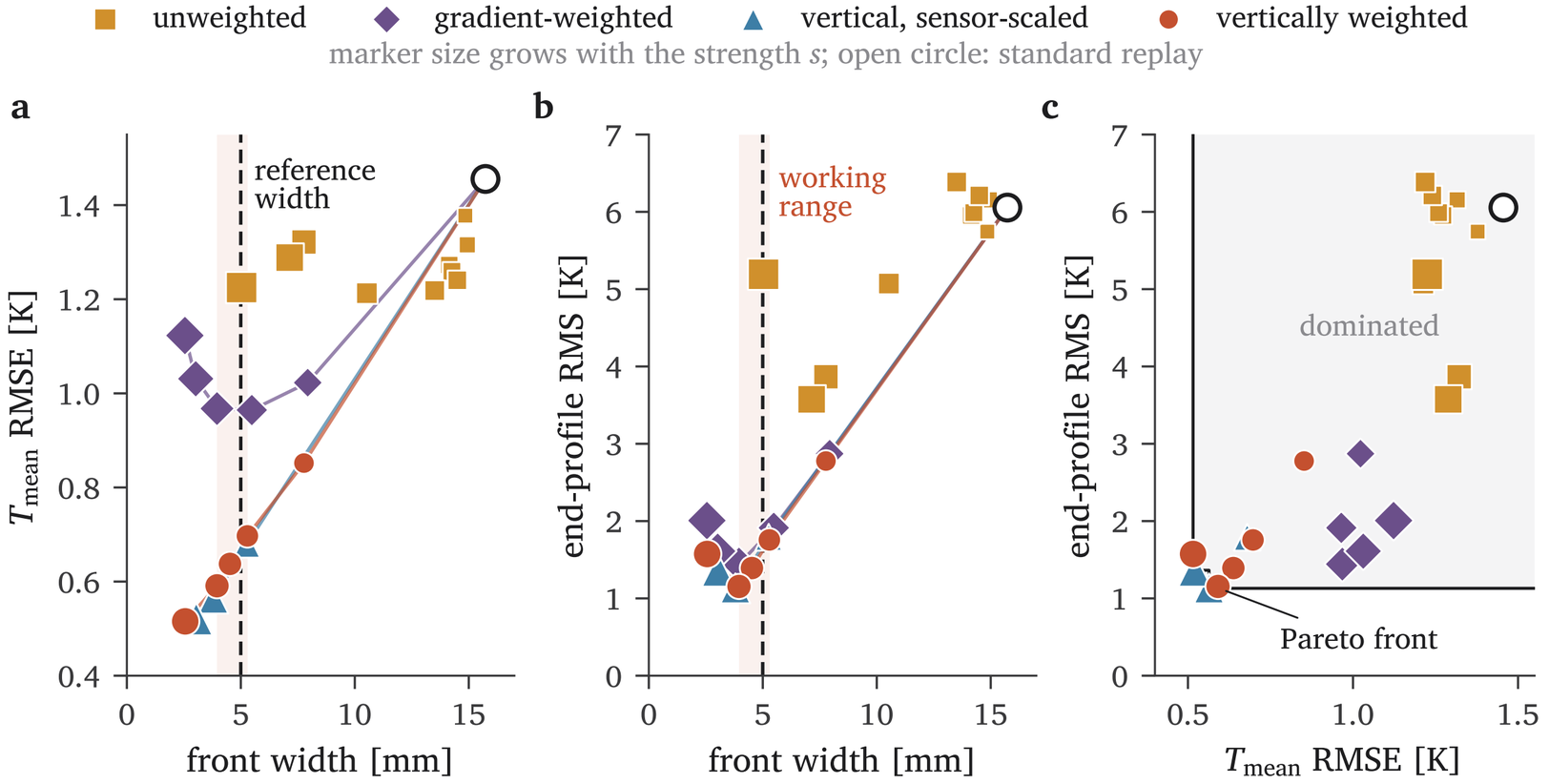
Reference: 569 min of solver time for 628 s of flow, image output removed

Replay: 0.81–4.73 min, mean 2.16 min for 600 s; grows with $f_{sd}$ (inner sweeps)

Ratio 120–700x across the sweep, about 260x on average, good to about 5 % (different spans)

The recording costs one full CFD run; the saving pays off from the second replay

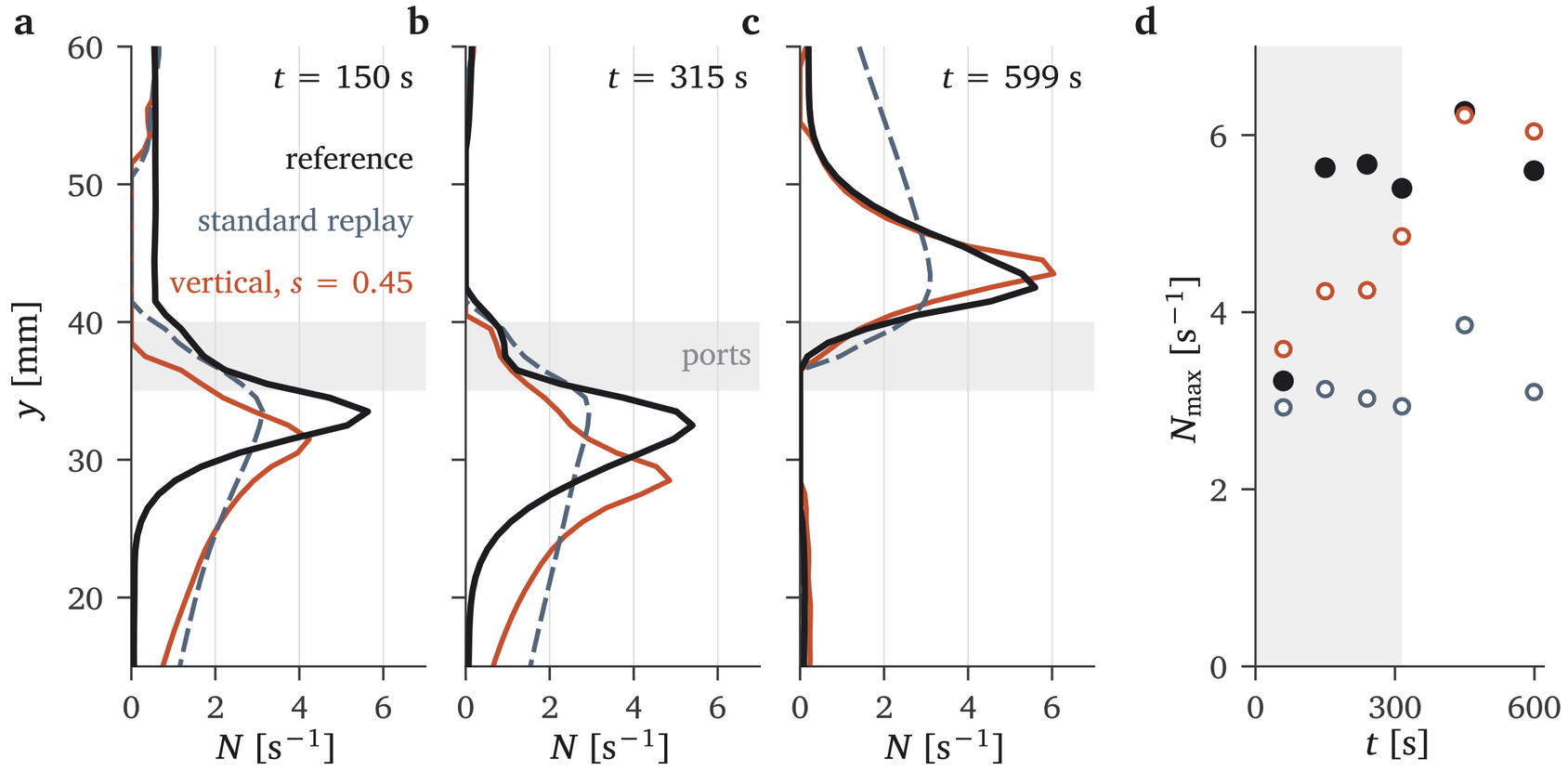
# The vertical form moves straight towards the reference width.




trace error, end profile and front width of every adiabatic correction run

THESIS FIG 5.6

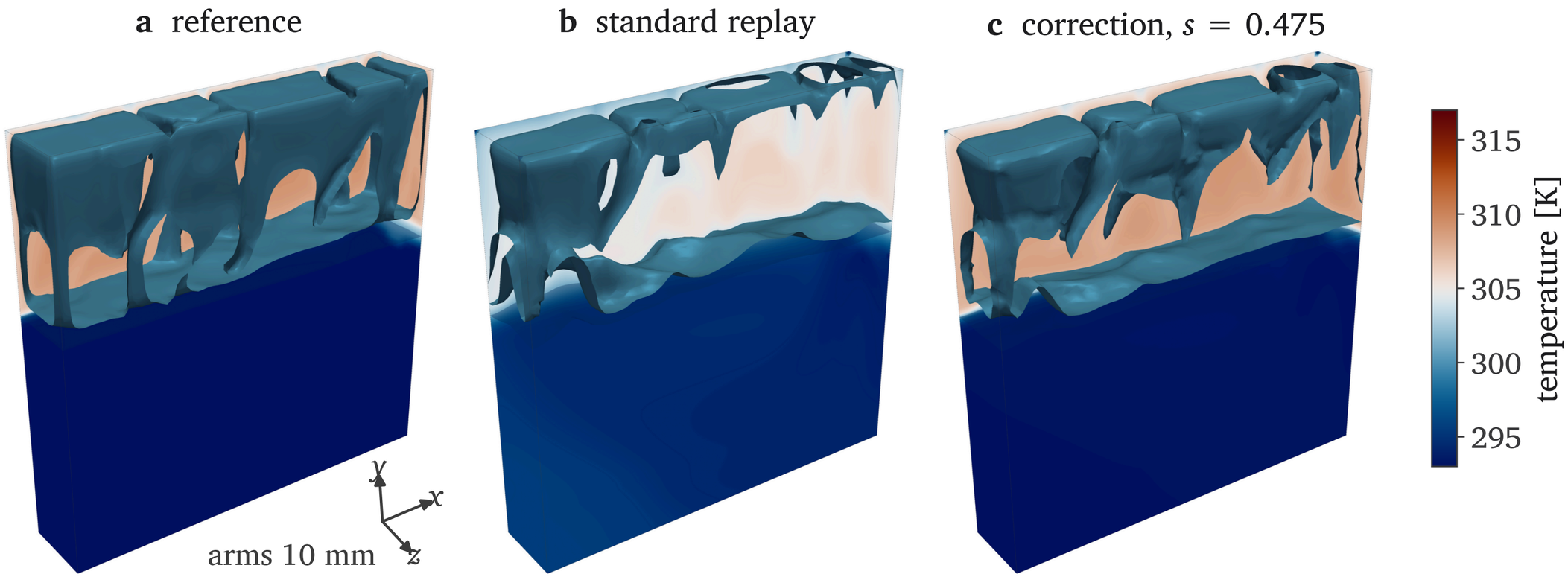
# The correction restores the stability of the front, not only its width.




N(y) across the front; peak: reference 5.4–6.3, standard 2.9–3.9, corrected 6.2 and 6.0 $s^{-1}$ at 450 and 599 s

THESIS FIG 5.7

# With the correction the recorded plumes reach the front again.




end state in 3D; the replay invents no motion, the plumes were in the maps

THESIS FIG 6.5

BACKUP · Q5

# The vessel study is running; no result is claimed here.

Two replays per feed position (lid-fed, floor-fed), with and without the correction, compared first with their own CFD and only then with thermocouple measurements.